

Public-key Cryptography Attacks Using Adiabatic Quantum Computer

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Abstract. We explore the application of the QUBO and Ising models to the integer factorization problem with implications for the security of public-key algorithms such as RSA. A key contribution is a program that applies existing algorithms to parameterize and simulate integer factorization through an Ising model in order to replicate previous works. Due to limited access to quantum hardware, we use classical heuristic methods to approximate solutions.

Keywords: Ising model · QUBO · integer factorization · quantum annealing · cryptography · adiabatic quantum computing.

1 Introduction

Cryptography is a fundamental technology for ensuring information security. Widely used public-key encryption algorithms, such as the RSA algorithm, rely on the computational difficulty of factoring a large integer of n bits into two prime factors, denoted as $\mathbf{N} = \mathbf{p} \cdot \mathbf{q}$.

If efficient methods were discovered to factor such integers, encryption schemes relying on the hardness of factorization would be at risk of compromise, potentially affecting critical sectors. Advancements in integer factorization research have a direct impact on the security of cryptographic systems.

On classical computers, Fermat’s factorization algorithm [2] can be used to decompose an integer. This algorithm is based on the difference of squares identity and seeks to express a large integer N as $N = a^2 - b^2 = (a - b)(a + b)$. It starts with $a = \lceil \sqrt{N} \rceil$ and incrementally increases a , checking at each step whether $a^2 - N$ is a perfect square. Once this condition is met, the factors of N can be recovered. This method is inefficient if the primes are far apart. The time complexity is $O(2^{n/4})$.

Shor’s algorithm [6] offers a quantum approach to integer factorization by transforming it into a period-finding problem. Theoretically, Shor’s algorithm can factor integers in quantum polynomial time, with a complexity of $O(n^3)$, while the best classical algorithms require exponential time with a complexity of $O(e^{\sqrt{n} \log n})$. However, current quantum computers do not yet have a sufficient number of qubits to factor large integers used in cryptographic applications.

A method proposed by Chinese researchers for quantum annealing-based integer factorization has gained widespread attention [7]. The technique [3] uses the optimization power of a quantum annealing to transform the problem of decomposing large integers into a quantum computational task of finding an energy ground state which is an optimal solution. The method can effectively decompose smaller integers under specific conditions, demonstrating the potential of quantum annealing in cryptanalysis.

In this article, we reformulate the factorization problem as an optimization task, which can be represented into Ising model. We develop a program capable of simulating integer factorization using heuristic methods to solve the problem on a classical computer. Although computation time on classical machines is longer than on quantum systems, this approach is necessary due to the limited availability and high cost of D-Wave quantum computers.

This paper firstly introduces the QUBO problem, Ising model and quantum annealing computer. Next, it applies the initial method to transform the problem into Ising model. Finally it presents the results and discussion and finishes with a conclusion.

Related work Many efforts have been made to solve the integer factorization problem using quantum or quantum-inspired methods.

Burges [1] laid the groundwork for quantum-inspired approaches by recasting factorization as an optimization problem, which enables its reformulation into a QUBO problem. This transformation allows the use of quantum annealers for solving such problems.

Jiang et al. [4] implemented integer factorization using D-Wave’s quantum annealer. Their approach involved encoding the problem into an Ising model and executing it on quantum hardware to find the solution.

Building upon these prior efforts, the present work proposes a framework that maps the integer factorization problem to an Ising model via QUBO reformulation. Unlike previous implementations, our approach simulates the solution process using classical heuristic solvers, thereby circumventing the technical limitations and high costs associated with current quantum annealing while offering a more accessible and flexible alternative for evaluating the feasibility of quantum-inspired factorization. We also draw the limits of such approach.

2 Models and Quantum Annealing

2.1 Models

QUBO (Quadratic Unconstrained Binary Optimization) is a mathematical optimization problem where the objective is to minimize a quadratic function with binary variables. The complexity of the classical QUBO problem is $2^n = O(\exp(n))$.

The standard form of a QUBO problem is to minimize $\sum_i \sum_j Q_{ij} x_i x_j$, where

- Q_{ij} is a matrix whose coefficients represent the interactions between variables x_i and x_j , with values in $\mathbb{R}$.
- The objective is to find $x = (x_1, x_2, \dots, x_n) \in \{0, 1\}^n$ in order to minimize the sum of the interactions between the variables.

Ising model comes from physics to describe the magnetism of ferromagnetic materials. It assumes that ferromagnetic materials consist of a set of spins (small magnetic needles) arranged in a regular pattern. Each spin can only take two orientations: up (+1) or down (−1). The spins interact with each other through energy constraints while being subjected to random disturbances from the environment, which can cause changes in orientation. The Ising models used in this article are only 2-dimensional: except for spins at the boundary, each spin has four neighbors: up, down, left, and right.

In the Ising model, we can construct V_N , a set of size N composed of vertices linked by an adjacency relation forming a $\mathbb{Z}^2$ lattice. Thus, there are N^2 vertices in V_N , and each vertex can be represented by a point with coordinates (x, y) when $1 \leq x, y \leq N$. Each vertex represents a spin numbered $i, i \in \{1, \dots, N^2\}$. Each spin has a value $\sigma_i \in \{-1, +1\}$, indicating its orientation and a *configuration* ω denotes the affectation of spins to the lattice.

The total energy of the system (or Hamiltonian) for a given configuration ω is expressed by $H(\omega) = -\sum_{i,j} J_{ij}\sigma_i\sigma_j - \sum_i h_i\sigma_i$, where

- J is real-valued; J_{ij} represents the interaction between spins i and j .
- $h_i \in \mathbb{R}$ is the interaction between the external field and spin i .

2.2 Quantum annealing

A key feature of the Quantum Adiabatic Computer is the tunnel effect [5], also known as **quantum tunneling**, a quantum mechanical phenomenon that allows a particle to pass through a potential barrier even if its energy is lower than the minimum energy required to overcome it according to classical physics. In quantum tunneling, the particle's wave function does not vanish at the barrier but instead decays within it, giving the particle a non-zero probability of appearing on the other side.

For combinatorial optimization problems, the adiabatic quantum computer uses quantum tunneling to escape local minima and find the global optimal solution.

3 Transformation into Ising model with the initial method

The integer factorization problem can be mapped to an Ising model, thereby providing a foundation for solving it using quantum annealing or other approximate algorithms. In this section, we focus on the initial modeling stage, where the integer factorization problem is first formulated as a QUBO problem and then converted into an Ising model in order to use a solver capable of finding the minimal energy.

3.1 Objective Function Construction

RSA factorization is to find two primes p and q such that $N = p \times q$. To formulate this as an optimization problem, we assume that the binary lengths of p and q are known.

To illustrate the concrete implementation of this method, we consider the integer $N = 21$ as an example, and demonstrate how the problem can be systematically transformed into an Ising expression. We assume that p has a bit length of 2 and q has a bit length of 3. Hence, they can be expressed as: $p = 2p_1 + 1$ and $q = 2q_1 + 4q_2 + 1$, where $p_1, q_1, q_2 \in \{0, 1\}$.

The objective of a QUBO problem is to minimize the sum of the interactions between variables, so we can define an objective function as:

$$f = (N - pq)^2 = [21 - (2p_1 + 1)(2q_1 + 4q_2 + 1)]^2 \quad (1)$$

The function achieves a minimum value of 0 at $pq = N$. Therefore, the problem can be reduced to an optimization problem of finding a configuration of variables in the space of Boolean variables that minimizes the objective function.

3.2 Polynomial Expansion and Degree Reduction

Since the variables take the values of 0 or 1, all the higher power terms can be transformed into linear terms and the QUBO formulation requires the objective function to be quadratic. To achieve this, we introduce auxiliary variables to eliminate higher-order terms. The formula defined in [4] in order to reduce the resulting 3-local terms to 2-local is while preserving its minimal energy is:

$$x_1 x_2 x_3 = x_4 x_3 + 2(x_1 x_2 - 2x_1 x_4 - 2x_2 x_4 + 3x_4) \text{ if } x_4 = x_1 x_2$$

After reducing higher-order terms to quadratic form and variable substitution, we obtain an expression in standard QUBO form, where all variables are binary and all terms are at most quadratic:

$$\begin{aligned} f = & 152p_1q_1 - 144p_1q_2 - 512p_1x_1 - 76p_1 + 16q_1q_2 \\ & - 512q_1x_1 - 76q_1 + 128q_2x_1 - 144q_2 + 768x_1 + 400 \end{aligned} \quad (2)$$

3.3 Mapping from QUBO to Ising Model

To transform a QUBO problem into Ising model, a common approach is to map variables. The conversion rule is given by the transformation f defined as:

$$\begin{aligned} f : \{0, 1\} &\longrightarrow \{-1, 1\} \\ x &\mapsto 2x - 1 \end{aligned}$$

Applying this transformation to the previous expression and simplifying the coefficients, we obtain the corresponding Ising formulation:

$$\begin{aligned} f = & 38p_1q_1 - 36p_1q_2 - 128p_1x_1 - 164p_1 + 4q_1q_2 - 128q_1x_1 \\ & - 124q_1 + 32q_2x_1 - 72q_2 + 160x_1 + 418 \end{aligned} \quad (3)$$

This method can also be extended to other fundamental number-theoretic problems in cryptography, such as the discrete logarithm problem [8], in both additive groups $(\mathbb{Z}_p, +)$ (as well as the group of rational points of an elliptic curve) and multiplicative groups $(\mathbb{Z}_p^*, \cdot)$, where one seeks a solution x such that $g \cdot x \equiv y \pmod{p}$ or $g^x \equiv y \pmod{p}$ (resp.).

These problems can similarly be formulated as Ising models and addressed using classical heuristic solvers, which is especially relevant to the security of schemes like Diffie–Hellman and elliptic curve cryptography.

4 Results and Discussion

Equation 3 provides the mathematical foundation and modeling framework for solving such problems using quantum annealers or approximate solvers. With the use of heuristic solver ³, we obtain a solution of the Ising model, which provides values for variables, from which p and q can be reconstructed.

The results given by the solver are sub-optimal. The solver was able to get the correct answer only when $N = 15, 21, 33, 39$. However, for values such as $N = 35$ and larger numbers, the solver fails to produce correct results. Analyzing the problem from a physical point of view, the reasons could be:

- In constructing the objective function, coupling terms for multiple spin variables are obtained after expanding equation 1. This process produces coupling coefficients of inconsistent sign, and a large number of coupling terms in the system may contradict each other, leading to a frustrated state, i.e., the inability to minimize all coupled energy terms simultaneously.
- The large differences in coupling strengths result in uneven energy scales. The product term generated by the high bits has a coefficient of $2^i \cdot 2^j$ in the energy function, resulting in an exponential numerical increase in the coupling strength. This dramatic imbalance makes the contribution of the higher bits to the system energy much larger than that of the lower bits. Different values of high bits will cause large energy differences between states. This discontinuity makes it easy for optimization methods based on slow energy reduction, such as annealing algorithms, to get stuck in local minima and not be able to jump out of them.

To enhance the accuracy of the solution:

- We can further reduce the solution space, e.g., reduce the complexity of the problem by minimizing the number of auxiliary variables in the process of optimizing higher-order terms to quadratic forms.
- We propose use the normalization of coupling coefficients or the weighted decay strategy, which effectively balances the effects of different variables on the energy function by scaling the various coefficients or introducing an exponential weighting function.

³ Dimod doc., https://test-projecttemplate-dimod.readthedocs.io/en/release_notes/index.html

- We can also adopt the method of local multiplication module, which divides the global multiplication process into multiple local computational blocks, each of which only deals with a limited range of variable couplings, thus constructing a structure characterized by local connectivity.

At <https://github.com/WWeishen/Factorization-Number/tree/main>, you may find our source code.

Conclusion

In this work, we investigated the integer factorization problem, which is a cornerstone of modern public-key cryptography, by reformulating the factorization task into a QUBO problem and subsequently mapping it onto the Ising model. Then we simulated the process using classical heuristic solvers.

We identified key challenges that hinder solution accuracy, including frustrated spin interactions, exponential disparity in coupling coefficients, and the difficulty of escaping local minima using annealing-based methods. To mitigate these issues, several enhancement strategies have been proposed, such as normalizing coupling strengths, employing weighted decay schemes, and decomposing global multiplication into locally connected modules.

While this approach does not yet scale, it lays the groundwork for future exploration of stronger quantum algorithms and hardware.

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